
Knowledge Distillation for Efficient On-Device Image Classification

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Code: <https://github.com/saadali18/on-device-image-classification>

Abstract

Deploying accurate image classifiers on resource-constrained edge devices requires compact models that preserve most of a large model’s accuracy. We study response-based knowledge distillation (Hinton et al., 2015) as a principled compression strategy on CIFAR-100, using a pretrained ResNet-110 teacher (73.48% Test Top-1) to supervise a family of smaller students via a combined hard-label cross-entropy and temperature-scaled KL divergence objective. Against a hard-label ResNet-20 baseline of 67.63% Test Top-1, our temperature sweep $\tau \in \{2, 4, 8, 16\}$ and loss-weighting sweep $\alpha \in \{0.1, 0.5, 0.9\}$ both exhibit clean inverted-U behaviour with peaks at $\tau^* = 8$ and $\alpha^* = 0.5$. An architectural-scaling sweep across depth (ResNet- $\{20, 32, 44, 56\}$) and width ($w \in \{0.5, 1, 2\}$) shows that distilled-student accuracy is monotonic in capacity in our regime: three of five capacity points exceed the teacher’s accuracy, with the wider ResNet-20 ($w = 2$, 1.10M parameters — still $1.6\times$ smaller than the teacher) reaching **75.43%** Top-1, **+7.80** pp above the hard-label baseline and **+1.95** pp above the teacher itself. The contrary failure mode appears at the low-capacity end: ResNet-20 narrowed to $w = 0.5$ (72k parameters) finishes 10.37 pp *below* baseline, demonstrating a minimum-capacity threshold below which dark knowledge cannot be absorbed. A distributional analysis visualises the teacher’s softened logits and a calibration analysis reveals a non-obvious side-effect: distillation degrades calibration (ECE rises from 6.23% to $\sim 10\%$) as the student inherits the teacher’s overconfidence pattern.

1. Introduction

Deep Convolutional Neural Networks (CNNs) like ResNet-110 achieve state-of-the-art accuracy in an over-parameterized regime (≈ 1.7 M parameters) (He et al., 2016),

but deployment to edge devices enforces hard constraints on memory and inference latency that such models violate. Buciluă et al. (2006) introduced model compression as a principled solution; Hinton et al. (2015) formalized this as *Knowledge Distillation* (KD) for neural networks.

In standard K -class classification, a network produces logits z_k passed through softmax

$$\hat{y}_k = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}}, \quad (1)$$

optimized via categorical cross-entropy against one-hot labels $\tilde{y}_i \in \{0, 1\}^K$:

$$L = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K y_{i,k} \log(\hat{y}_{i,k}). \quad (2)$$

Hard labels discard inter-class similarity. A large teacher network¹ trained to near-convergence absorbs this structure into its output distribution $\tilde{y}^T$, placing small but nonzero probabilities on semantically related classes — *dark knowledge* (Hinton et al., 2015). KD incorporates $\tilde{y}^T$ directly into the student’s training objective, exposing this richer supervision signal. This project studies that transfer: *what* the teacher transfers, how hyperparameters control it, and where it fails.

2. Literature Review

2.1. Model Compression Landscape

Model compression falls into three families (Gou et al., 2021): (i) *Pruning* removes weights or filters from a pre-trained network (Cheng et al., 2024); (ii) *Quantization* reduces numerical precision (e.g. FP32 $\rightarrow$ INT8) to shrink memory and accelerate arithmetic (Nagel et al., 2021); (iii) *Distillation* trains a new compact student network supervised by a larger teacher. All three are mutually compatible and frequently pipelined. This paper is scoped to response-based distillation, which transfers knowledge through the teacher’s output distribution.

¹We use superscripts T and S to denote teacher and student quantities respectively.

2.2. Response-Based Distillation

The hard-label gradient problem. The gradient of Eq. (2) with respect to the student’s pre-softmax logit z_k^S is

$$\frac{\partial L}{\partial z_k^S} = \hat{y}_k^S - y_k. \quad (3)$$

For every non-target class $y_k = 0$, so the update becomes $z_k^S \leftarrow z_k^S - \eta \hat{y}_k^S$, strictly pushing non-target logits toward $-\infty$ regardless of class similarity.

Temperature scaling. Hinton et al. (2015) introduced the temperature-scaled softmax: for teacher logits $\tilde{z}^T$,

$$\hat{y}_k^{T,\tau} = \frac{\exp(z_k^T/\tau)}{\sum_{j=1}^K \exp(z_j^T/\tau)}, \quad (4)$$

applied equally to the student to produce $\hat{y}_k^{S,\tau}$. Increasing $\tau > 1$ softens the distribution, amplifying the relative magnitude of non-target class gradients and exposing inter-class similarity to the student. At $\tau \rightarrow \infty$ the distribution approaches uniform; at $\tau \rightarrow 0$ it collapses to a one-hot argmax.

Training objective. The student optimizes a weighted sum of a hard-label term,

$$L_{\text{hard}} = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K y_{i,k} \log \hat{y}_{i,k}^{S,1}, \quad (5)$$

and a soft-label KL divergence term,

$$L_{\text{soft}} = \frac{1}{N} \sum_{i=1}^N D_{\text{KL}}(\hat{y}_i^{T,\tau} \parallel \hat{y}_i^{S,\tau}). \quad (6)$$

Expanding and grouping by entropy terms gives

$$L_{\text{soft}} = H(\hat{y}^{T,\tau}, \hat{y}^{S,\tau}) - H(\hat{y}^{T,\tau}).$$

Since the teacher is frozen, $H(\hat{y}^{T,\tau})$ is constant w.r.t. the student, so minimizing L_{soft} is equivalent to minimizing the cross-entropy $H(\hat{y}^{T,\tau}, \hat{y}^{S,\tau})$.

The τ^2 scaling factor. We defer the full gradient derivation to Appendix A; the result is $\partial L_{\text{soft}}/\partial z_k^S = (z_k^S - z_k^T)/(K\tau^2)$, which scales as $1/\tau^2$ relative to the hard-label gradient. Without correction, raising τ shrinks the soft gradient as $1/\tau^2$, reverting the student to hard-label training; multiplying L_{soft} by τ^2 cancels this shrinkage. The combined objective is:

$$L_{\text{KD}} = \alpha \cdot L_{\text{hard}} + (1 - \alpha) \cdot \tau^2 \cdot L_{\text{soft}} \quad (7)$$

where $\alpha \in [0, 1]$ balances hard vs. soft supervision and τ controls distribution softness.

3. Methodology

Dataset. All experiments use CIFAR-100: 60,000 colour images at 32×32 across $K = 100$ classes (50k train / 10k test).

Teacher. ResNet-110 (He et al., 2016) is trained from scratch on CIFAR-100 with standard cross-entropy for 160 epochs (LR milestones $\{80, 120\}$), then its weights are frozen for all distillation experiments. The longer schedule ensures the teacher converges to a high accuracy ceiling before being used as a soft-target source.

Student. ResNet-20 (He et al., 2016) ($\approx 0.27\text{M}$ parameters, $6.3\times$ smaller than the teacher) serves as the default student. It is architecturally homogeneous with ResNet-110, isolating depth as the primary capacity variable.

3.1. Baseline and Student Variants

We train ResNet-20 from scratch with standard CCE (no teacher) to establish the lower bound for distillation to beat. For the capacity study we define two families: a *depth* series ResNet- d , $d \in \{20, 32, 44, 56\}$ (width 16), and a *width* series ResNet-20 with multiplier $w \in \{0.5, 1, 2\}$ (base channels $\{8, 16, 32\}$). The ResNet-20 ($w = 1$) serves as the shared pivot.

3.2. Training Configuration

Table 1. **Training configuration.** Teacher uses 160 epochs with milestones $\{80, 120\}$; all else identical.

Hyperparameter	Value
Optimiser	SGD + Momentum (0.9)
Initial LR	0.1 (MultiStep)
LR milestones / γ	epochs 60, 80 / 0.1
Weight decay	5×10^{-4}
Batch size / Epochs	128 / 100
Loss	Cross-entropy

4. Results

4.1. Baseline and Teacher

Training ResNet-20 from scratch on CIFAR-100 establishes the lower bound; the ResNet-110 teacher uses the longer schedule in Section 3. Best-checkpoint metrics are in Table 2. The $73.48 - 67.63 = 5.85$ pp teacher–baseline gap defines the headroom available to distillation.

Table 2. Baseline and teacher best-checkpoint metrics.

Metric	Baseline (ResNet-20)	Teacher (ResNet-110)
Parameters	278,324	1,736,564
Best Epoch	89	129
Test Top-1	67.63%	73.48%
Test Top-5	91.31%	92.74%

4.2. Hyperparameter Ablation

Temperature. Fixing $\alpha = 0.5$, we sweep $\tau \in \{2, 4, 8, 16\}$ (Table 3). The sweep produces a clean inverted-U peaking at $\tau^* = 8$, recovering 40% of the teacher-baseline gap at zero parameter cost.

 Table 3. Temperature ablation ($\alpha = 0.5$, ResNet-20, 100 epochs).

τ	Test Top-1	Δ vs. baseline
2	69.49%	+1.86
4	69.64%	+2.01
8	69.98%	+2.35
16	69.73%	+2.10

Loss weighting. Using $\tau^* = 8$, we sweep $\alpha \in \{0.1, 0.5, 0.9\}$ (Table 4). All three values improve over the baseline; the balanced $\alpha^* = 0.5$ is optimal, with $\alpha = 0.9$ recovering only 62% of the gain — confirming the soft-label term does real work.

 Table 4. Loss-weighting ablation ($\tau = 8$, ResNet-20, 100 epochs).

α	Test Top-1	Δ vs. baseline
0.1	69.74%	+2.11
0.5	69.98%	+2.35
0.9	69.07%	+1.44

4.3. Architectural Scaling

Using $(\tau^*, \alpha^*) = (8, 0.5)$, we measure distillation benefit along depth and width axes.

Depth. Table 5 sweeps ResNet- d , $d \in \{20, 32, 44, 56\}$ at width $w = 1$.

Width. Table 6 sweeps $w \in \{0.5, 1, 2\}$ at depth 20.

 Table 5. Depth ablation ($\tau = 8$, $\alpha = 0.5$, 100 epochs). Δ_T : gap vs. teacher (73.48%).

Student	Params	Top-1	Δ base	Δ_T
ResNet-20	278,324	69.98%	+2.35	-3.50
ResNet-32	472,756	72.39%	+4.76	-1.09
ResNet-44	667,188	73.52%	+5.89	+0.04
ResNet-56	861,620	74.72%	+7.09	+1.24

 Table 6. Width ablation (ResNet-20, $\tau = 8$, $\alpha = 0.5$, 100 epochs). Δ_T : gap vs. teacher.

w	Params	Top-1	Δ base	Δ_T
0.5	71,756	57.26%	-10.37	-16.22
1.0	278,324	69.98%	+2.35	-3.50
2.0	1,096,196	75.43%	+7.80	+1.95

5. Dark Knowledge Analysis

Figure 2 shows, for six held-out test images, the teacher’s softmax distribution at $T = 1$ and $T = 4$.² At $T = 1$ the teacher’s distribution is near one-hot; at $T = 4$ it spreads tens of percentage points of mass onto semantically related classes. This off-argmax structure — *dark knowledge* (Hinton et al., 2015) — tells the student not just *which* class is correct but *how* the teacher’s representation organises the class manifold, explaining the accuracy gains from distillation.

6. Calibration Analysis

Table 7 reports Expected Calibration Error (ECE, 15 bins) and mean max-confidence $\bar{p}_{\max}$ on the CIFAR-100 test set; reliability diagrams are in Figure 3.

Table 7. Calibration on CIFAR-100 test set (ECE, 15 bins).

Model	Top-1	ECE	$\bar{p}_{\max}$
Teacher (ResNet-110)	73.48%	12.54%	0.860
Hard-label student	67.63%	6.23%	0.738
Distilled ($\tau=8$, $\alpha=0.5$)	69.98%	10.32%	0.803
Distilled ($w=2$, best)	75.43%	9.97%	0.854

Counter-intuitively, the hard-label student (ECE 6.23%) is better calibrated than both distilled students despite lower Top-1 accuracy. The KL term transfers the teacher’s full output distribution—including its overconfidence profile ($\bar{p}_{\max} = 0.860$)—so the distilled student’s confidence shifts toward the teacher’s, raising ECE to $\sim 10\%$. Accuracy and calibration are not jointly improved by KD; post-hoc cal-

²Training used $\tau^* = 8$; $T = 4$ gives a more legible bar plot with identical top- k ordering.

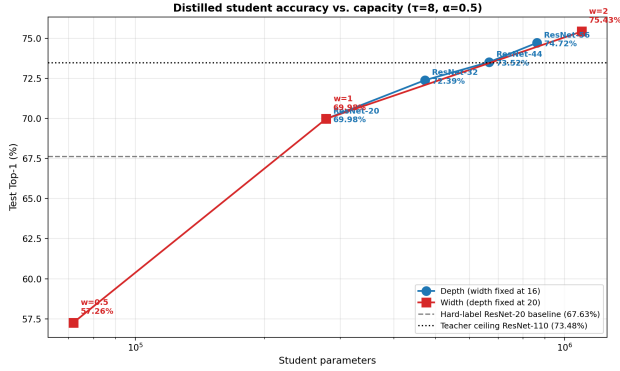


Figure 1. Distilled accuracy vs. capacity ($\tau = 8$, $\alpha = 0.5$). Blue circles: depth series; red squares: width series. Gray/black lines: baseline and teacher.

ibration is required for probability-sensitive downstream tasks.

7. Discussion

Temperature traces a clean inverted-U. The τ sweep confirms KD theory: too-sharp distributions approach hard labels while too-flat distributions approach uniform, both eliminating discriminative signal. $\tau^* = 8$ exceeds Hinton’s canonical $\tau \approx 4$ because our ResNet-110 teacher retains $> 40\%$ of mass on its argmax at $\tau = 4$, requiring further softening to expose useful inter-class structure; with 100 classes, non-target logit gradients are also inherently small and benefit from the additional amplification.

α is less sensitive but still optimal at the balanced setting. All three α values improve over the baseline (+1.44–+2.35 pp), but the inverted-U peaks at $\alpha^* = 0.5$. The $\alpha = 0.9$ deterioration (only 62% of peak gain) rules out a pure-regularisation explanation, confirming the soft-label term is supplying genuinely new information. Conversely, $\alpha = 0.1$ remains competitive, showing ground-truth anchoring retains value even when the teacher dominates.

Capacity scaling: “student exceeds teacher”. Monotone scaling holds throughout our regime: three of five students cross the teacher’s 73.48% ceiling because our teacher sits below the ResNet-CIFAR family ceiling ($\sim 76\%$), so its soft-label signal continues to carry useful regularization well into the high-capacity regime. The headline result is **75.43% Top-1** (+7.80 pp) at 1.10M parameters — still $1.6\times$ smaller than the teacher. The $w = 0.5$ (72k param) cell demonstrates the lower-end failure mode (−10.37 pp below baseline): the student lacks capacity to absorb either the dark-knowledge or discriminative signal, and the mixed distillation loss becomes noisier than CE alone.

Accuracy and calibration are not jointly improved. Distillation degrades calibration (ECE rises from 6.23% to 10.32%) even as it improves Top-1 by up to +7.80 pp. The KL term transfers the teacher’s full output distribution—including its overconfidence profile—not merely the argmax. Applications requiring calibrated probabilities must pair KD with a post-hoc calibration step (e.g. temperature scaling on a validation split). Single-seed training and a below-ceiling teacher (73.48% vs. literature $\sim 76\%$) are the primary limitations of this study.

8. Conclusion

We studied response-based knowledge distillation (Hinton et al., 2015) on CIFAR-100, using a ResNet-110 teacher (73.48% Top-1) to supervise a $6.3\times$ smaller ResNet-20 student. Our principal findings:

- Distillation closes 40% of the teacher–baseline gap at zero parameter cost.** The best distilled ResNet-20 ($\tau^* = 8$, $\alpha^* = 0.5$) reaches 69.98%, recovering 2.35 of the 5.85 pp gap.
- Temperature traces an inverted-U peaking at $\tau^* = 8$.** This validates the theoretical prediction that too-sharp or too-flat distributions both lose discriminative signal; $\tau^* = 8 > 4$ (Hinton) because our teacher has sharper logits and CIFAR-100 has 100 classes.
- Loss weighting is less sensitive but still favours $\alpha^* = 0.5$.** All α values improve over baseline; $\alpha = 0.9$ recovers only 62% of peak gain, ruling out a pure-regularisation effect.
- Distilled students can exceed the teacher when capacity is sufficient.** Three of five Step-3 students surpass the teacher’s ceiling, with the best ($w = 2$) reaching **75.43%** at 1.10M params; the $w = 0.5$ cell (72k params) falls 10.37 pp below baseline.
- Distillation degrades calibration even as it improves accuracy.** ECE rises from 6.23% (baseline) to 10.32% as the KL term transfers the teacher’s overconfidence profile; post-hoc calibration is required for probability-sensitive applications.
- Dark knowledge is directly observable.** Temperature-softened teacher distributions reveal inter-class similarity structure, and distilled student outputs are visibly closer to these distributions than to the hard-label baseline.

Future work includes cross-architecture distillation (CNN to MobileNetV2), feature-based methods (Romero et al., 2015), and combining KD with pruning or quantization.

Teacher dark knowledge (T=tau) is closer to the distilled student than to the hard-label baseline.

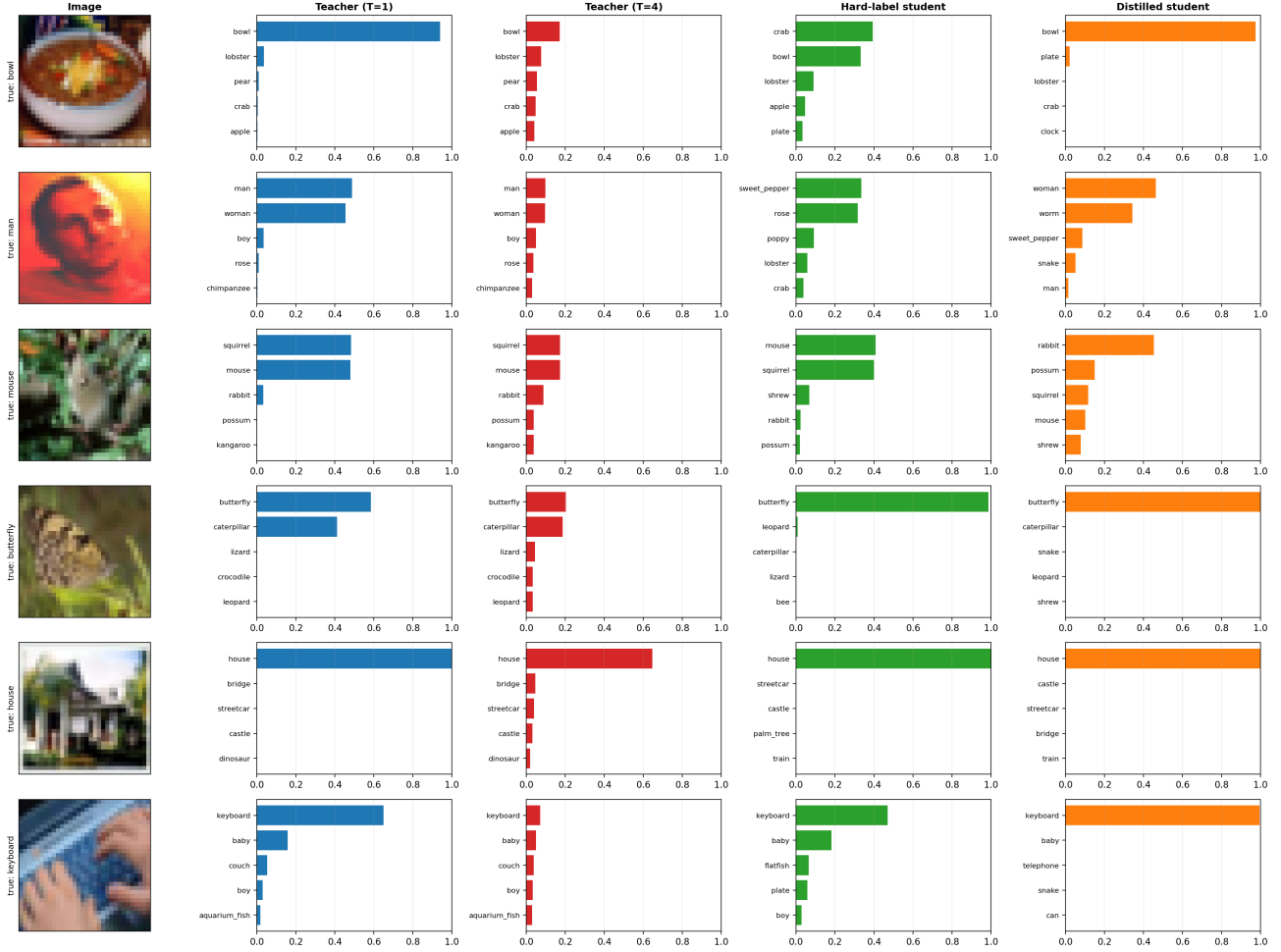


Figure 2. **Teacher dark knowledge.** Each row: input image; teacher softmax at $T=1$; teacher softmax at $T=4$; hard-label baseline student; distilled student. The softened teacher (red) reveals the inter-class similarity structure transferred by the distillation loss.

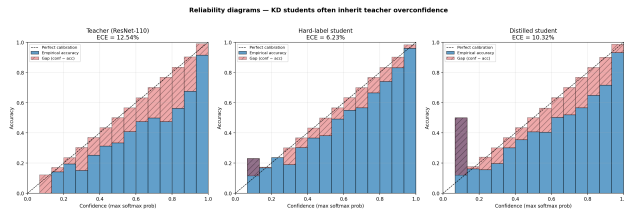


Figure 3. **Reliability diagrams (15 bins).** The hard-label baseline is the best-calibrated model; distilled students inherit the teacher’s overconfidence.

Statement on the Use of AI Tools

Claude Code was used for software scaffolding, debugging, and structural report drafts. All experimental design, results interpretation, and mathematical derivations were produced by the team.

A. Gradient of L_{soft} with Respect to z_k^S

Hinton et al. (2015) argue that the soft-label loss in combined distillation loss must be weighted by τ^2 . To prove this to be the case, we differentiate L_{soft} , defined in Eq. (6), with respect to a student logit z_k^S . The gradient for a single sample (dropping index i for convenience) is, by chain rule,

$$\frac{\partial L_{\text{soft}}}{\partial z_k^S} = \sum_{j=1}^K \frac{\partial L_{\text{soft}}}{\partial \hat{y}_j^{S,\tau}} \cdot \frac{\partial \hat{y}_j^{S,\tau}}{\partial z_k^S}. \quad (8)$$

We evaluate each term individually. Since the teacher terms are constant with respect to θ_S , applying the logarithmic quotient rule and differentiating $-\sum_j \hat{y}_j^{T,\tau} \log \hat{y}_j^{S,\tau}$ with respect to $\hat{y}_j^{S,\tau}$ gives

$$\frac{\partial L_{\text{soft}}}{\partial \hat{y}_j^{S,\tau}} = -\frac{\hat{y}_j^{T,\tau}}{\hat{y}_j^{S,\tau}}. \quad (9)$$

The student probability $\hat{y}_j^{S,\tau}$ depends on z_k^S only through the scaled pre-activation $a_k^S = z_k^S/\tau$, so a further application of the chain rule gives

$$\frac{\partial \hat{y}_j^{S,\tau}}{\partial z_k^S} = \frac{\partial \hat{y}_j^{S,\tau}}{\partial a_k^S} \cdot \underbrace{\frac{\partial a_k^S}{\partial z_k^S}}_{=1/\tau}, \quad (10)$$

where, by applying the quotient rule and considering the two distinct cases (diagonal and off-diagonal elements) for the jacobian,

$$\frac{\partial \hat{y}_j^{S,\tau}}{\partial a_k^S} = \hat{y}_j^{S,\tau} (\delta_{jk} - \hat{y}_k^{S,\tau}). \quad (11)$$

Combining Eqs. (10) and (11) and substituting it along with Eq. (9) into Eq. (8)

$$\begin{aligned} \frac{\partial L_{\text{soft}}}{\partial z_k^S} &= \sum_{j=1}^K \left(-\frac{\hat{y}_j^{T,\tau}}{\hat{y}_j^{S,\tau}} \right) \cdot \frac{1}{\tau} \hat{y}_j^{S,\tau} (\delta_{jk} - \hat{y}_k^{S,\tau}) \\ &= \frac{1}{\tau} \sum_{j=1}^K \left(-\hat{y}_j^{T,\tau} \right) (\delta_{jk} - \hat{y}_k^{S,\tau}) \\ &= \frac{1}{\tau} \left[-\hat{y}_k^{T,\tau} + \hat{y}_k^{S,\tau} \sum_{j=1}^K \hat{y}_j^{T,\tau} \right] \\ &= \frac{1}{\tau} (\hat{y}_k^{S,\tau} - \hat{y}_k^{T,\tau}) \end{aligned}$$

Substituting Eq. (4):

$$= \frac{1}{\tau} \left(\frac{\exp(z_k^S/\tau)}{\sum_{j=1}^K \exp(z_j^S/\tau)} - \frac{\exp(z_k^T/\tau)}{\sum_{j=1}^K \exp(z_j^T/\tau)} \right)$$

Applying first-order Taylor approximation $e^u \approx 1 + u$, valid when τ is large relative to the logit magnitudes (Hinton et al., 2015):

$$\begin{aligned} &\approx \frac{1}{\tau} \left(\frac{1 + z_k^S/\tau}{\sum_{j=1}^K (1 + z_j^S/\tau)} - \frac{1 + z_k^T/\tau}{\sum_{j=1}^K (1 + z_j^T/\tau)} \right) \\ &= \frac{1}{\tau} \left(\frac{1 + z_k^S/\tau}{K + \frac{1}{\tau} \sum_{j=1}^K z_j^S} - \frac{1 + z_k^T/\tau}{K + \frac{1}{\tau} \sum_{j=1}^K z_j^T} \right) \end{aligned}$$

Assuming logits are zero-meaned, i.e. $\sum_{j=1}^K z_j^S = \sum_{j=1}^K z_j^T = 0$ (Hinton et al., 2015):

$$\begin{aligned} &= \frac{1}{\tau} \left(\frac{1 + z_k^S/\tau}{K} - \frac{1 + z_k^T/\tau}{K} \right) \\ &= \frac{1}{K\tau} \left(1 + \frac{z_k^S}{\tau} - 1 - \frac{z_k^T}{\tau} \right) \\ &= \frac{1}{K\tau^2} (z_k^S - z_k^T). \end{aligned}$$

The gradient scales as $1/\tau^2$, justifying the τ^2 prefactor in Eq. (7): without it, raising τ would exponentially suppress the soft-label gradient and revert training to hard labels.

The full derivation intermediate steps (Eqs. (8)–(9)–(11)) follow from combining the logarithmic quotient rule, the quotient rule for the softmax Jacobian (diagonal and off-diagonal cases), and the chain rule through $a_k^S = z_k^S/\tau$.

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